

Automobile Sales Analysis Dashboard Report



1. Introduction

The Automobile Sales Analysis Dashboard was developed using Microsoft Power BI to analyze automobile sales performance and provide actionable business insights. The objective of this project was to transform raw sales data into meaningful visualizations that help stakeholders monitor sales performance, profitability, customer trends, regional performance, and product contribution.

The dashboard enables interactive analysis through slicers, charts, KPIs, and advanced Power BI visualizations, allowing users to make informed business decisions based on data-driven insights.

2. Requirement Analysis

Business Requirements

The project aimed to answer the following business questions:

- What is the total sales generated?
- Which regions contribute the highest sales?
- Which vehicle categories perform best?
- How does sales vary over time?
- What is the profit generated from sales?
- Which products contribute most to revenue?
- Are sales targets being achieved?
- What factors influence sales performance?

Tools Used

- Microsoft Power BI
- DAX (Data Analysis Expressions)
- Microsoft Excel

Dataset Details

The dataset contains automobile sales information including:

- Vehicle Type
- Model
- Make
- Country
- Sales Amount
- Cost Price
- Delivery Charges
- Discounts

- Labor Cost
- Spare Parts Cost
- Invoice Date

3. Methodology

Data Preparation

- Imported Excel dataset into Power BI.
- Verified data quality and structure.
- Removed inconsistencies and validated data types.
- Created relationships where required.

Data Modeling

Created calculated measures using DAX:

- Total Sales
- Profit
- Profit Margin
- Average Sales
- Running Total Sales
- Year-To-Date Sales
- Sales Growth %
- Customer Count

Dashboard Development

Created interactive dashboards using:

- KPI Cards
- Line Charts
- Bar Charts
- Pie Charts
- Scatter Charts
- Gauge Charts
- Treemaps

- Matrix Visuals
- Maps
- Histograms
- Decomposition Trees

4. Dashboard Components

KPI Cards

Displayed:

- Total Sales
- Profit
- Profit Margin %
- Total Customers
- YTD Sales

Purpose:

Provides a quick overview of business performance.

Monthly Sales Trend (Line Chart)

Purpose:

Analyze monthly sales performance and identify seasonal trends.

Insights:

- Identified periods of sales growth.
- Highlighted months with lower sales performance.

Sales vs Profit Analysis (Combo Chart)

Purpose:

Compare sales and profitability over time.

Insights:

- High sales periods do not always generate maximum profit.
- Profitability depends on operational costs and discounts.

Sales by Region (Map/Pie Chart)

Purpose:

Analyze geographic distribution of sales.

Insights:

- Certain regions contributed significantly higher revenue.
- Regional differences indicate market opportunities.

Treemap – Sales Contribution by Vehicle Type

Purpose:

Measure contribution of each vehicle category.

Insights:

- High-performing vehicle categories were identified.
- Product concentration became visible.

Top 5 Selling Models (Bar Chart)

Purpose:

Identify best-performing vehicle models.

Insights:

- Top-selling models contribute significantly to overall revenue.
- Useful for inventory and marketing planning.

Scatter Plot – Sales vs Profit

Purpose:

Study the relationship between sales and profitability.

Insights:

- Identified highly profitable products.
- Revealed products with high sales but lower margins.

Gauge Chart – Sales Target Achievement

Purpose:

Compare actual sales against predefined targets.

Insights:

- Measures performance against business goals.
- Provides immediate performance visibility.

Matrix Visual

Purpose:

Display sales information across regions and vehicle types.

Insights:

- Enables detailed comparison.
- Supports drill-down analysis.

Histogram

Purpose:

Display distribution of sales amounts.

Insights:

- Shows common sales ranges.
- Helps understand sales behavior patterns.

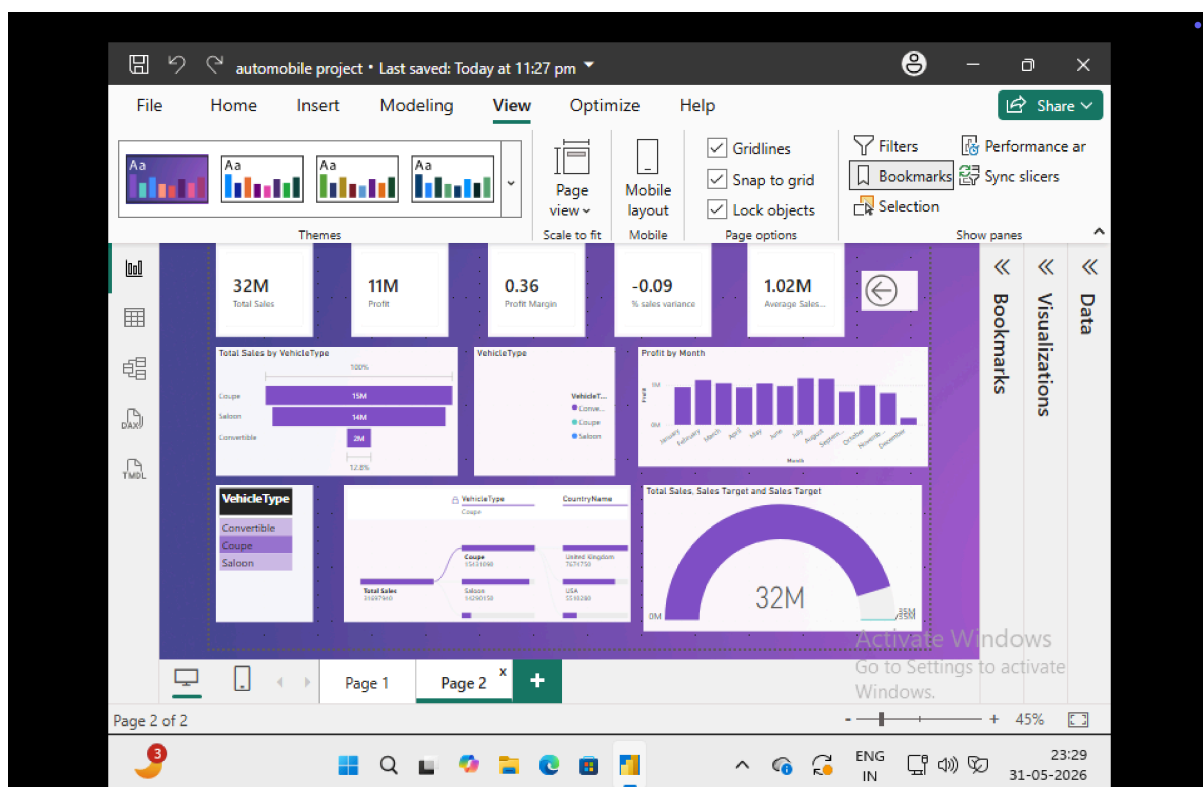
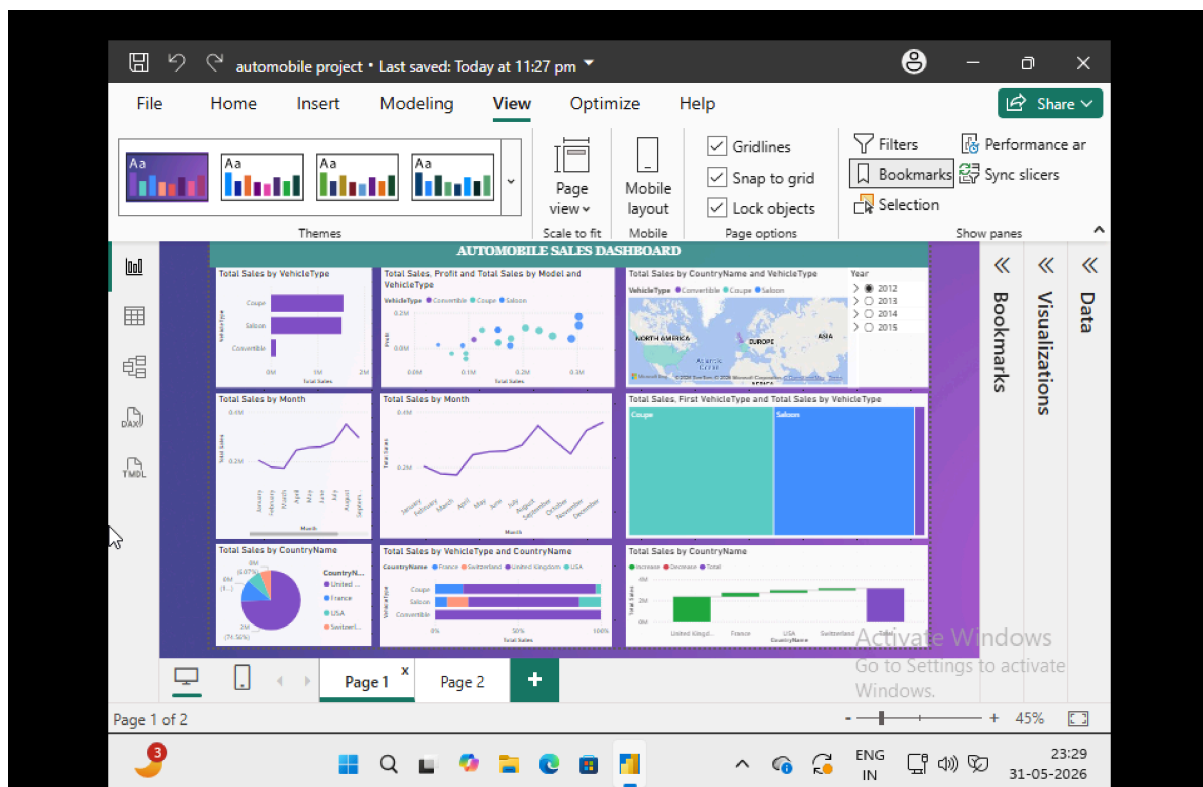
Decomposition Tree

Purpose:

Analyze factors contributing to total sales.

Insights:

- Identified key drivers of sales.
- Allowed drill-down from country to vehicle type and model.



5. Key Findings

- Total sales generated strong business revenue.
- Certain regions consistently outperformed others.
- Specific vehicle categories contributed the majority of sales.
- Top-performing models accounted for a significant share of revenue.
- Profitability varied despite similar sales volumes.
- Sales trends showed identifiable growth periods.
- Customer purchasing behavior varied by region and vehicle type.

6. Actionable Insights

Sales Strategy

Focus marketing investments on top-performing regions.

Product Strategy

Increase inventory for high-demand vehicle categories.

Profitability Improvement

Review discount policies and operational costs for low-margin products.

Customer Growth

Target underperforming regions with promotional campaigns.

Performance Monitoring

Use KPI and Gauge visuals to continuously track business objectives.

7. Approaches Used

Descriptive Analytics

Used to summarize historical sales performance.

Comparative Analysis

Compared regions, products, and monthly performance.

Trend Analysis

Studied sales growth over time.

Performance Analysis

Measured achievement against targets.

Root Cause Analysis

Used decomposition trees to identify key sales drivers.

8. Conclusion

The Automobile Sales Analysis Dashboard successfully transformed raw sales data into meaningful business intelligence. Through interactive visualizations, DAX calculations, and advanced analytics, the dashboard provides comprehensive insights into sales performance, profitability, customer behavior, and regional contribution.

The solution enables stakeholders to monitor KPIs, identify opportunities, improve profitability, and make data-driven decisions. The project demonstrates effective use of Microsoft Power BI for business intelligence and analytics applications.